BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/Desktop/Cornell_Courses/FA_25/BEE_4750/hw5-xindicliu`
```

```julia
using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF**: Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF**: Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE**: Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

Answer:

Quantity	Formula	Actual Calculation	Value
Overall recycling fraction	$r = \sum w_i r_i$	$0.15(0) + 0.40(0.55) + ... + 0.03(0)$	0.3775 (37.75%)
Overall MSW ash fraction	$a_{\text{MSW}} = \sum w_i a_i$	$0.15(0.08) + 0.40(0.07) + ... + 0.03(0.70)$	0.1641 (16.41%)

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Answer:

Let:

$ x_{i,j} = $ mass flow (Mg/day) from city i to facility j, where $i \in \{1,2,3\}$ and $j \in \{LF, MRF, WTE\}$.

$ y_{MRF,k} = $ mass flow (Mg/day) of residue from MRF to facility k, where $k \in \{LF, WTE\}$.

$ y_{WTE,LF} = $ mass flow (Mg/day) of ash from WTE to LF.

$ _j = $ binary variable indicating whether facility j is used, $j \in \{LF, MRF, WTE\}$, with $\delta_j \in \{0,1\}$.

All mass-flow variables are nonnegative.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Answer:

Parameter definitiosn: | Parameter | Meaning | |———-|———-| | t | transport cost ($\$/\text{Mg} \cdot \text{km}$) | | $d_{i,j}$ | distance from city i to facility j | | c_j^{tip} | tipping fee at facility j | | c^{rec} | recycling processing cost (\$/Mg recycled) | | r_{overall} | overall recycling fraction (=0.3775) | | r_{MRF} | MRF recycling rate (=0.40) | | F_j | fixed daily cost for facility j | | C_j | capacity of facility j | | M_j | big-M value for facility j (typically $M_j = C_j$) |

The objective is to minimize the total daily system cost:

$$Z = C_{trans} + C_{tip} + C_{rec} + C_{fixed}.$$

where - transport cost from cities to facilities:

$$C_{trans,city} = t \sum_{i=1}^{3} \sum_{j \in \{LF,MRF,WTE\}} d_{i,j}\, x_{i,j}.$$

- transport cost for inter-facility flows:

$$C_{trans,inter} = t(d_{MRF,LF}\, y_{MRF,LF} + d_{MRF,WTE}\, y_{MRF,WTE} + d_{WTE,LF}\, y_{WTE,LF}).$$

1. Total transport cost

$$C_{trans} = C_{trans,city} + C_{trans,inter}.$$

2. Tipping fees

$$C_{tip} = c_{LF}^{tip}(\sum_{i=1}^{3} x_{i,LF}+y_{MRF,LF}+y_{WTE,LF})+c_{MRF}^{tip}(\sum_{i=1}^{3} x_{i,MRF})+c_{WTE}^{tip}(\sum_{i=1}^{3} x_{i,WTE}+y_{MRF,WTE}).$$

3. MRF recycling costs

$$C_{rec} = c^{rec}\, r_{\mathrm{MRF}}(\sum_{i=1}^{3} x_{i,MRF}).$$

4. Fixed facility operating costs

$$C_{fixed} = F_{LF}\,\delta_{LF} + F_{MRF}\,\delta_{MRF} + F_{WTE}\,\delta_{WTE}.$$

Putting all components together:

$$Z = C_{trans} + C_{tip} + C_{rec} + C_{fixed}.$$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Answer:

City mass balance (for $i = 1, 2, 3$):

$$\sum_{j\in\{LF,MRF,WTE\}} x_{i,j} = S_i.$$

MRF inflow and split:

$$X_{MRF} = \sum_{i=1}^{3} x_{i,MRF}.$$

$$M^{rec} = r_{\mathrm{MRF}}\, X_{MRF}, \quad M^{res} = (1 - r_{\mathrm{MRF}})\, X_{MRF}.$$

MRF residual routing:

$$y_{MRF,LF} + y_{MRF,WTE} = (1 - r_{\mathrm{MRF}}) \sum_{i=1}^{3} x_{i,MRF}.$$

WTE inflow:

$$X_{WTE} = \sum_{i=1}^{3} x_{i,WTE} + y_{MRF,WTE}.$$

WTE ash:

$$y_{WTE,LF} = 0.1641 \sum_{i=1}^{3} x_{i,WTE} + 0.1386\, y_{MRF,WTE}.$$

Landfill inflow:

$$X_{LF} = \sum_{i=1}^{3} x_{i,LF} + y_{MRF,LF} + y_{WTE,LF}.$$

Capacity constraints:

$$\sum_{i=1}^{3} x_{i,MRF} \leq C_{MRF},$$

$$\sum_{i=1}^{3} x_{i,WTE} + y_{MRF,WTE} \leq C_{WTE},$$

$$\sum_{i=1}^{3} x_{i,LF} + y_{MRF,LF} + y_{WTE,LF} \leq C_{LF}.$$

Big-M linking constraints:

(to enforce zero flow when facility closed)

$$\sum_{i=1}^{3} x_{i,MRF} \leq M_{MRF}\, \delta_{MRF},$$

$$\sum_{i=1}^{3} x_{i,WTE} + y_{MRF,WTE} \leq M_{WTE}\, \delta_{WTE},$$

$$\sum_{i=1}^{3} x_{i,LF} + y_{MRF,LF} + y_{WTE,LF} \leq M_{LF}\, \delta_{LF}.$$

- Choose $M_j \geq C_j$, typically $M_j = C_j$.

Nonnegativity and integrality:

$$x_{i,j} \geq 0,\ y_{MRF,LF} \geq 0,\ y_{MRF,WTE} \geq 0,\ y_{WTE,LF} \geq 0.$$

$$\delta_{LF}, \delta_{MRF}, \delta_{WTE} \in \{0, 1\}.$$

Problem 1.5

Find the optimal solution (using `JuMP` to solve the problem). Report the optimal objective value.

```
# Data
S = [100.0, 90.0, 120.0]
C_LF = 200.0; C_MRF = 350.0; C_WTE = 210.0

# MRF recycling rate = 37.75%
r_MRF = 0.3775

# Ash fractions from problem 1.1
a_MSW = 0.1641
a_res = 0.1386

# Transportation cost ($/Mg-km)
t = 1.5

# Distances (km)
d_city_LF = [5.0, 15.0, 13.0]
d_city_MRF = [30.0, 25.0, 45.0]
d_city_WTE = [15.0, 10.0, 20.0]
d_MRF_LF = 32.0; d_MRF_WTE = 15.0; d_WTE_LF = 18.0

# Tipping fees ($/Mg)
c_tip_LF = 50.0; c_tip_MRF = 7.0; c_tip_WTE = 60.0

# Recycling cost ($/Mg recycled)
c_rec = 40.0

# Fixed costs ($/day)
F_LF = 2000.0; F_MRF = 1500.0; F_WTE = 2500.0

# Big-M values (choose tight = capacities)
M_LF = C_LF; M_MRF = C_MRF; M_WTE = C_WTE

# Model
model = Model(HiGHS.Optimizer)
set_silent(model)

# Variables
@variable(model, x[i = 1:3, j = 1:3] >= 0)
```

```julia
@variable(model, y_MRF_LF >= 0)
@variable(model, y_MRF_WTE >= 0)
@variable(model, y_WTE_LF >= 0)
@variable(model, delta_LF, Bin)
@variable(model, delta_MRF, Bin)
@variable(model, delta_WTE, Bin)

# City mass balance constraints
@constraint(model, [i = 1:3], sum(x[i, j] for j = 1:3) == S[i])

# MRF residue split for non-recycled portion
@constraint(model, y_MRF_LF + y_MRF_WTE == (1 - r_MRF) * sum(x[i, 2] for i = 1:3))

# WTE ash
@constraint(model, y_WTE_LF == a_MSW * sum(x[i, 3] for i = 1:3) + a_res * y_MRF_WTE)

# Capacity constraints
@constraint(model, sum(x[i, 1] for i = 1:3) + y_MRF_LF + y_WTE_LF <= C_LF)
@constraint(model, sum(x[i, 2] for i = 1:3) <= C_MRF)
@constraint(model, sum(x[i, 3] for i = 1:3) + y_MRF_WTE <= C_WTE)

# Big-M linking constraints
@constraint(model, sum(x[i, 1] for i = 1:3) + y_MRF_LF + y_WTE_LF <= M_LF * delta_LF)
@constraint(model, sum(x[i, 2] for i = 1:3) <= M_MRF * delta_MRF)
@constraint(model, sum(x[i, 3] for i = 1:3) + y_MRF_WTE <= M_WTE * delta_WTE)

# Objective: Minimize total cost
transport_city = t * (
    sum(d_city_LF[i] * x[i, 1] for i = 1:3) +
    sum(d_city_MRF[i] * x[i, 2] for i = 1:3) +
    sum(d_city_WTE[i] * x[i, 3] for i = 1:3)
)

transport_inter = t * (d_MRF_LF * y_MRF_LF + d_MRF_WTE * y_MRF_WTE + d_WTE_LF * y_WTE_LF)

tip_cost = c_tip_LF * (sum(x[i, 1] for i = 1:3) + y_MRF_LF + y_WTE_LF) +
           c_tip_MRF * (sum(x[i, 2] for i = 1:3)) +
           c_tip_WTE * (sum(x[i, 3] for i = 1:3) + y_MRF_WTE)

recycle_cost = c_rec * r_MRF * sum(x[i, 2] for i = 1:3)

fixed_cost = F_LF * delta_LF + F_MRF * delta_MRF + F_WTE * delta_WTE
```

```julia
@objective(model, Min, transport_city + transport_inter + tip_cost + recycle_cost + fixed_cos

# Solve
optimize!(model)

# Output results
status = termination_status(model)
println("Status: ", status)

if status == OPTIMAL
    println("Objective value: \$", round(objective_value(model); digits = 2))

    println("\nFlows (x[i, j]) where j = 1:LF, 2:MRF, 3:WTE:")
    for i in 1:3, j in 1:3
        v = value(x[i, j])
        if v > 1e-6
            println("x[$i, $j] = ", round(v; digits = 3))
        end
    end

    println("\ny_MRF_LF = ", round(value(y_MRF_LF); digits = 3))
    println("y_MRF_WTE = ", round(value(y_MRF_WTE); digits = 3))
    println("y_WTE_LF = ", round(value(y_WTE_LF); digits = 3))

    println("\ndelta_LF  = ", Int(value(delta_LF)))
    println("delta_MRF = ", Int(value(delta_MRF)))
    println("delta_WTE = ", Int(value(delta_WTE)))
end
```

```
Status: OPTIMAL
Objective value: $27855.48

Flows (x[i, j]) where j = 1:LF, 2:MRF, 3:WTE:
x[1, 1] = 100.0
x[2, 3] = 90.0
x[3, 1] = 78.405
x[3, 3] = 41.595

y_MRF_LF = 0.0
y_MRF_WTE = 0.0
y_WTE_LF = 21.595
```

```
delta_LF  = 1
delta_MRF = 0
delta_WTE = 1
```

Answer:

The optimal solution is WTE takes all the 90 Mg of wastes from City 2, then another 41.595 Mg from City 3. It will generate about 21.595 Mg ashes that will then be transported to LF. More than the ashes, LF also takes 100 Mg of wastes from City 1 and 78.405 Mg of wastes from City 3. This gives MTE 131.595 Mg of total wastes to treat, and 200 Mg afterwards in total for LF, while MRF is completely unused. The optimal objective value is $27855.48.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Answer:

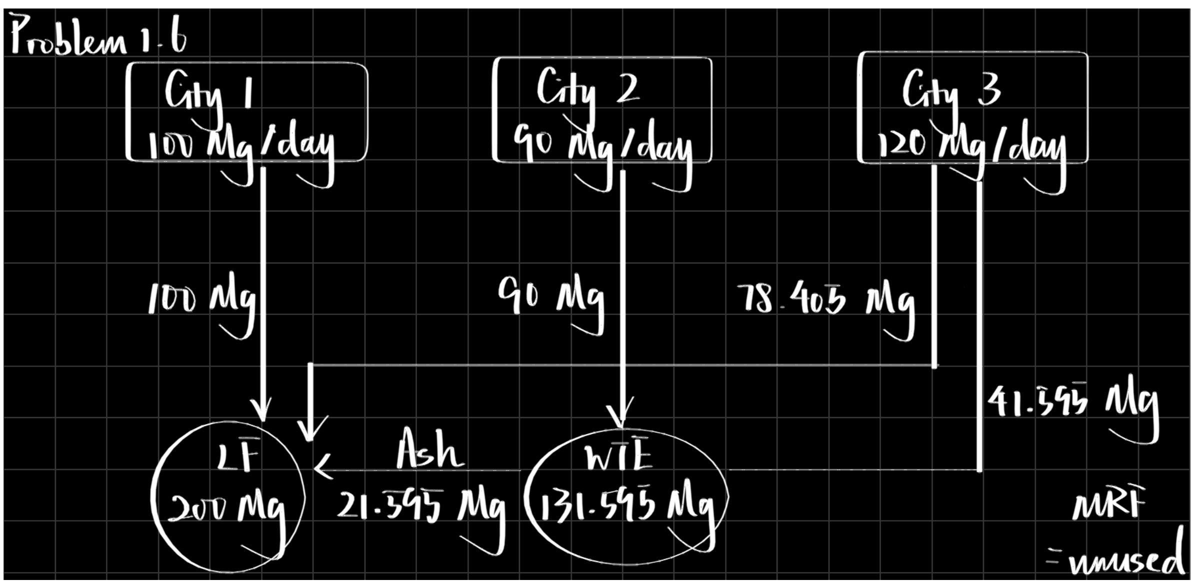

Figure 1: Description

The facility that will not be used is the MRF. This solution makes sense because using the MRF incurs not only a fixed daily cost of $1500, but also a processing cost of $40/Mg recylced. With a low tipping fee of $7/Mg, the savings from sending waste through MRF versus other alternatives is small, so the combined costs outweigh the savings here. In addition, with the LF capacity available up to 200 Mg and WTE available up to 210 Mg, the model can meet demand without paying the MRF costs. Thus, not using MRF is economically and physically consistent given the data, with capacity limits satisfied, mass conserved, and objective minimized.

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100$MW. In period 2, the demand is projected to be $d_2 = 1200$MW, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Answer:

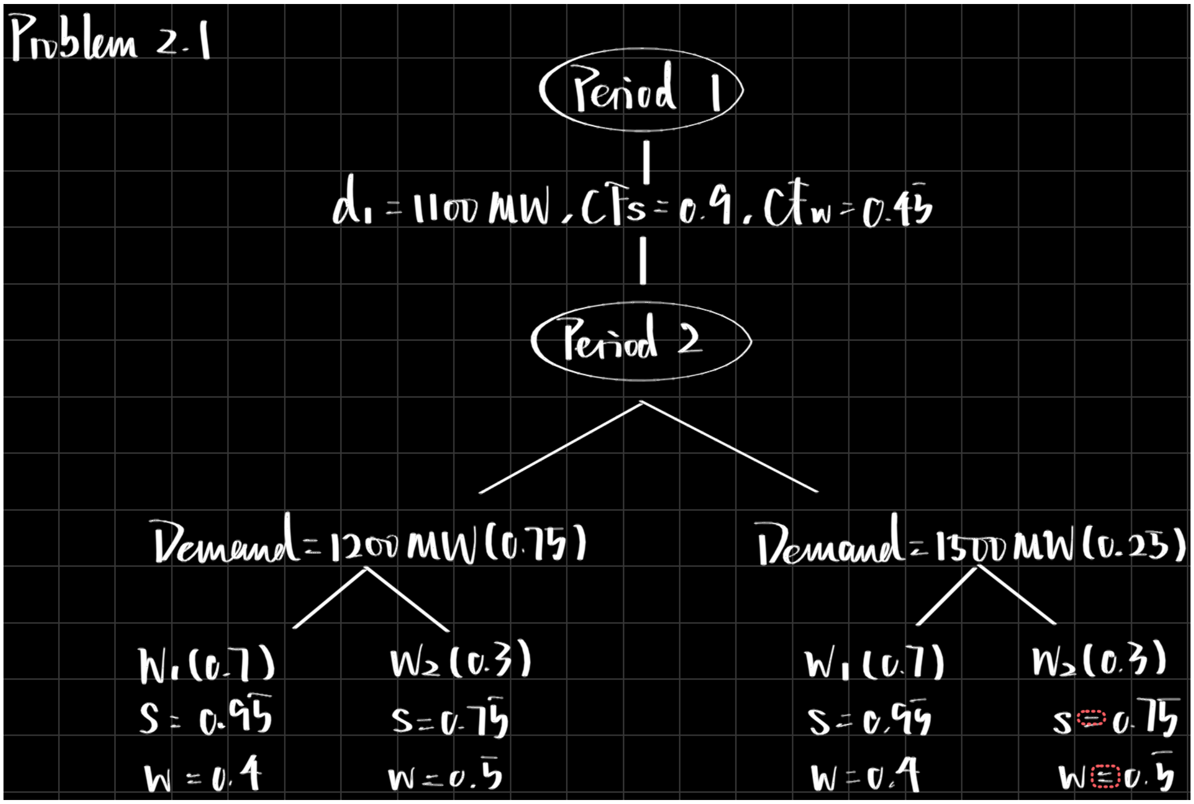

Figure 2: Description

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

Answer:

Sets: - Generators $i \in \mathcal{G}$. - Time periods $t = 1, 2$. - Scenarios $s \in \mathcal{S} = \{1, 2, 3, 4\}$.

Parameters: - $P_i^{\min}, P_i^{\max}$ = min/max output of generator i. - c_i = variable cost of generator i. - R_i = ramp limit of generator i. - $d_1 = 1100$. - $d_2^s \in \{1200, 1500\}$ depending on scenario. - Solar/wind capacity factors: - Period 1: $\mathrm{CF}_{\text{solar}}^{(1)} = 0.9,\ \mathrm{CF}_{\text{wind}}^{(1)} = 0.45$. - Period 2: $\mathrm{CF}_{\text{solar}}^{s},\ \mathrm{CF}_{\text{wind}}^{s}$. - Scenario probabilities: - $p_1 = 0.525,\ p_2 = 0.225,\ p_3 = 0.175,\ p_4 = 0.075$.

Decision variables: - $g_{i,1}$ = generation of unit i in period 1. - $g_{i,2}^s$ = generation of unit i in period 2 under scenario s.

Objective:

$$\min \sum_i c_i g_{i,1} \; + \; \sum_s p_s \sum_i c_i g_{i,2}^s$$

Constraints:

1. Period-1 capacity limits:
$$P_i^{\min} \leq g_{i,1} \leq P_i^{\max}.$$

2. Period-1 renewable availability:
$$g_{\text{wind},1} \leq 0.45\, P_{\text{wind}}^{\max}, \qquad g_{\text{solar},1} \leq 0.90\, P_{\text{solar}}^{\max}.$$

3. Period-1 demand:
$$\sum_i g_{i,1} = 1100.$$

4. Period-2 capacity limits:
$$P_i^{\min} \leq g_{i,2}^s \leq P_i^{\max} \qquad \forall s.$$

5. Period-2 renewable availability:
$$g_{\text{wind},2}^s \leq \mathrm{CF}_{\text{wind}}^s\, P_{\text{wind}}^{\max}, \qquad g_{\text{solar},2}^s \leq \mathrm{CF}_{\text{solar}}^s\, P_{\text{solar}}^{\max} \qquad \forall s.$$

6. Period-2 demand (scenario-dependent):
$$\sum_i g_{i,2}^s = d_2^s \qquad \forall s.$$

7. Ramping:
$$-R_i \leq g_{i,2}^s - g_{i,1} \leq R_i \qquad \forall i, s.$$

8. Nonnegativity:
$$g_{i,1} \geq 0, \qquad g_{i,2}^s \geq 0.$$

References

List any external references consulted, including classmates.

Used ChatGPT for problems 1.3, 1.4, 2.2, for the purpose of helping with LaTex syntax, since it is really complicated to utilize with all the different symbols and versions. OpenAI. (2025). ChatGPT (Feb 1 GPT-4 model) [Large language model]. https://chat.openai.com